- Input at least one operational spacecraft and one debris/object record.
- Propagate both objects over a chosen analysis window.
- Compute distance-vs-time, TCA, dCA and relative velocity.
- Create an uncertainty ensemble and visualize the Future State Envelope.
- Run No Action plus two candidate Δv scenarios.
- Repeat across multiple objects and 24 h / 7 d / 30 d horizons.
- Compute project-defined FEI and decision-sensitivity metrics.
- Store run metadata so every screenshot/result is reproducible.
- Build a validation mode separate from the presentation dashboard.

16. Validation Plan — The Evidence Judges Can Trust

Test	Method	Evidence
Propagation accuracy	Compare against a trusted/reference implementation for selected cases/	Position/state difference table
Time-step convergence	Repeat with finer time resolution	Convergence plot
Closest approach	Check that the reported minimum is stable under refinement	dCA/TCA comparison
Zero-maneuver test	$\Delta v = 0$ must reproduce No Action	Scenario equality test
Uncertainty convergence	Increase Monte Carlo sample count	Distribution stability plot
Uncertainty behavior	Increase assumed input uncertainty	Envelope should respond consistently
Threat-transfer test	Use a controlled multi-object scenario	Before/after object table
FEI weight test	Vary weights and inspect result stability	Sensitivity chart
Decision sensitivity	Repeat across uncertainty/parameter sweeps	Scenario frequency chart
End-to-end test	Run a complete mission case from data ingestion to dashboard	Run ID + screenshots

Do not invent accuracy numbers. Fill the results only after running the implementation. The research dossier should show both successful and failed/edge cases where useful.

17. Digital 'Photo Proof' / Evidence Pack

Because the organizer identifies this as a software track, physical machine parts are not required for the core solution. Strong evidence is digital and reproducible:

- Screenshot of the raw public/simulated orbital data used.
- Screenshot of the running backend/API and a real response.
- Cesium 3D Earth with actual propagated tracks.
- Distance-vs-time plot with TCA marker.
- Closest-approach numeric card.
- Monte Carlo uncertainty cloud / Future State Envelope.
- Scenario branch: No Action vs A vs B.
- Multi-object threat-transfer visualization.
- 24 h / 7 d / 30 d exposure comparison.
- FEI and decision-sensitivity outputs generated by the live code.
- Validation graphs and test-run identifiers.
- GitHub/project structure and selected source-code modules.

QR-code strategy

Do not add a fake 'scanner' component to the engineering architecture. Instead, place a QR code on the final presentation/dossier that links to the live prototype, GitHub repository, technical R&D; dossier or demo video. The judge's phone becomes the scanner; the QR code is an evidence gateway, not part of the orbital sensor system.

18. Research Questions for the Team to Answer Before Submission

Question	Expected evidence
Which orbital data source and epoch did we use?	Raw input + retrieval date
Which propagator/implementation is used?	Library/version + method note
How is TCA found?	Algorithm description + convergence test
What uncertainty model is assumed?	Parameters/covariance description
How many Monte Carlo samples?	Run configuration + convergence result
What exactly is our risk index?	Formula + calibrated/controlled tests
What exactly is FEI?	Formula + explicit 'project-defined' label
How are maneuvers represented?	Execution time + Δv vector + propagation procedure
Can a maneuver increase another threat?	Multi-object experiment
Does the decision change under uncertainty?	Sensitivity experiment
What is not flight-qualified?	Explicit limitations slide

19. Recommended Research Dossier Structure

1. Official problem statement • 2. Debris environment • 3. Observation vs prediction • 4. Existing conjunction workflow • 5. Gap after the alert • 6. Proposed architecture • 7. Data sources • 8. Propagation method • 9. Close approach • 10. Uncertainty • 11. Future State Envelope • 12. Scenario engine • 13. Multi-object analysis • 14. Multi-horizon • 15. Threat transfer • 16. FEI • 17. Decision sensitivity • 18. Explainability • 19. Validation • 20. Real-world case studies • 21. Implementation evidence • 22. Limitations • 23. Future work • 24. References • 25. QR links.

20. What We Should NOT Claim

• Do not claim to have invented orbital tracking, conjunction assessment, Monte Carlo, covariance modeling or orbital propagation.

• Do not call FEI an industry-standard metric.

• Do not present illustrative FEI/decision-sensitivity numbers as measured results.

• Do not call the software a physical debris scanner or radar.

• Do not claim flight-qualified maneuver optimization.

• Do not claim autonomous spacecraft control.

• Do not claim an operational reconstruction of a real collision unless the event is actually reconstructed and validated with appropriate data.

21. Reference Starting Points

The following sources are appropriate starting points for the team's literature/data collection. Because this PDF is generated without live web verification in this chat, the team should open each source and record the exact access date, document/version and specific page or dataset used before final submission.

Source	Use
Hackathon organizer — ProblemStatements_Aerospace_Hackathon_2026	Authoritative competition requirements for Problem Statement 03.
NASA Orbital Debris Program Office (ODPO)	Orbital debris environment, historical events and technical background.
ESA Space Debris Office / Space Environment	European debris environment and space-safety context.
CelesTrak	Public orbital element/catalog data and TLE/GP-related resources.
Space-Track.org	Official U.S. government orbital-data service; access/terms apply.
NASA conjunction / ISS safety communications	Public operational context for conjunction assessment and avoidance.
Peer-reviewed astrodynamics / conjunction-assessment literature	Methods for uncertainty, TCA, collision risk and propagation validation.

22. Final Research Narrative

Physical layer: sensors and tracking networks observe objects. **Data layer:** observations are turned into orbital states or public orbital elements. **Prediction layer:** propagation estimates where objects may be in the future. **Conjunction layer:** relative geometry identifies close approaches. **Decision layer:** our prototype adds uncertainty-aware future-state exploration, candidate scenario comparison, multi-object and multi-horizon consequences, threat transfer, FEI and decision sensitivity. **Human layer:** an operator reviews the modeled consequences; the prototype does not autonomously command a spacecraft.

The strongest demonstration is therefore not a fake physical scanner. It is a reproducible chain from real/public orbital data → physics-based propagation → close-approach calculation → uncertainty ensemble → scenario comparison → validation evidence. That directly addresses the organizer's software-track requirement while giving the project a deeper decision-support contribution.

FROM ONE PREDICTED FUTURE → TO PLAUSIBLE FUTURES

Explore plausible futures. Understand the consequences. Support better decisions.